Stage-Wise Attribution of Geometric Error in Automatic Anatomical Surface Reconstruction From Computed Tomography

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***Abstract*—Automatic pipelines turn computed tomography (CT) into surface models, but their geometric error is rarely traced to its source, and prior error budgets add stage errors measured one at a time. This study crosses resolution, segmentation source and reconstruction stages in a factorial design against two references: four analytic phantoms with exact surfaces (7,760 meshes), and expert-labelled CT of the spleen and, held out, the pancreas (41 scans each), resampled before segmentation by TotalSegmentator in fast and full-resolution modes (49,200 meshes). Spacing had the largest descriptive share of error variance on exact surfaces (38.8%) and on correct segmentations (spleen 50.5%, pancreas 74.2%). Simulated coarsening moved a correct surface by ASSD $\approx 0.177\,\Delta$, with Δ the sampling added; fitted on the spleen, it predicted the pancreas within 0.030 mm. For the engine, between-scan variation dominated (89.5% and 99.9%); its fast-mode spleen error was unchanged from native to 3 mm grids, but not for the pancreas ($+0.120$ mm) or in full-resolution mode ($+0.420$ mm), and 5 mm slices raised it throughout. Of six hypotheses fixed before the pancreas was processed, four held, one partly. Decimation broke closure without moving unsmoothed geometry, and meshing on the engine's 3 mm grid restored it.**

***Index Terms*—Marching cubes, mesh decimation, Taubin smoothing, surface distance, digital phantoms, slice thickness, segmentation evaluation, 3D printing.**

I. Introduction

Surface models built from CT are now routine in surgical planning, teaching and 3D printing [1], and engines such as TotalSegmentator [2], built on nnU-Net [3], have made segmentation automatic. A pipeline still makes several decisions after segmentation: it extracts an isosurface [4], [5], smooths it [6] and simplifies it [7]. Each can move the surface, and the scan's resolution limits what any of them can recover.

Accuracy studies of such models come mainly from 3D printing, where total error is split into segmentation, digital editing and printing error [1]. Their methodology measures each stage separately, on one phantom or structure with threshold-based segmentation, and adds the partial errors [8]; automatic engines, meanwhile, are evaluated at mask level [2], [9], so the surface a user finally sees is rarely scored. Two methodological gaps follow: an additive error budget cannot represent interactions between stages, and without an exact reference the error of the reference itself cannot be separated from that of the pipeline.

This paper attributes geometric error in *automatic* pipelines, against exact and expert references, through five questions. **Q1:** How much error does reconstruction itself introduce when the true surface is known? **Q2:** How far does simulated coarser acquisition alone move the surface of a correct segmentation, and how well can the shift be predicted? **Q3:** How does an engine's surface error depend on resolution and engine mode, and how is its variance shared between scans, resolution and reconstruction? **Q4:** How do deployed settings affect geometry, topology and local defects? **Q5:** Do the spleen findings hold on a held-out organ, under criteria fixed before its data were analysed?

The contributions are methodological first: (1) a two-reference factorial design that crosses the tested reconstruction stages with two segmentation sources, an expert mask and one engine in two modes, with resolution controlled before segmentation; (2) a scan-clustered, descriptive attribution in which spacing carries the largest share of error variance for correct segmentations and between-scan variation for the engine, re-tested on a held-out organ against pre-specified criteria; (3) an empirical sampling relationship with prediction intervals, validated across scans, levels, a second slice profile and a second organ; and (4) practical findings on Dice, blur units, decimation, meshing grids and local defects. The study makes no clinical claim; its anatomy is two abdominal organs from one public challenge, and its coarsening is simulated. A supplement (S1–S14) holds the extended analyses.

II. Related Work

Error decomposition in anatomical modelling. Schulze *et al.* reviewed 139 studies and organised their errors into segmentation, digital editing and printing error: the median segmentation error was 0.8 mm against 0.26 mm for printing, no average was found for digital editing, and the total did not exceed the partial errors, which may compensate each other [1]. Juergensen *et al.* evaluated the three stages individually on one physical phantom, varying slice thickness, kernel, threshold, smoothing, software and printer, and computed the total error as their sum [8]. Ahmed *et al.* crossed slice thickness and pixel size on six cadaver mandibles, scored against surface scans of the dissected bone, and found that slice thickness alone mattered [10].

Reconstruction and evaluation. Marching cubes [4], [5], Taubin smoothing [6] and decimation [7] are standard; smooth-

TABLE I
DESIGN OF THE CLOSEST PRIOR STUDIES AND OF THIS WORK (PRIOR DESIGNS AS STATED IN THEIR ABSTRACTS).

Study	Reference	Segmentation	Design	Scale
Juergensen [8]	3D scan of a physical phantom	threshold	stages one at a time; total = sum	1 phantom
Ahmed [10]	scan of dissected bone	not stated	slice × pixel size	6 cadavers
Schulze [1]	(review)	various	categorised reported errors	139 studies
This work	exact surfaces; expert masks	deep-learning engine (2 modes); expert mask	full factorial; scan-clustered; held-out organ	4 phantoms; 41 + 41 scans

ing trades staircase artefacts against shape and volume on medical surfaces [11], and simplification error is usually measured between a mesh and its own original [12], not against anatomy. Guidelines recommend boundary metrics alongside overlap [13], [14], yet benchmarks score masks, not meshes [9], and quality estimation without ground truth [15], [16] likewise stops at the mask.

How this study differs. Table I compares the designs; four differences matter methodologically. First, segmentation is a deep-learning engine, in two modes, evaluated as one pipeline stage, with the expert mask passed through the same pipeline as a segmentation-free control. Second, exact analytic surfaces measure reconstruction error, including the floor of the reference procedure itself. Third, stages are crossed in a full factorial design, so interactions are measured: on phantoms they carry about a third of the variance and reverse the effect of smoothing between shapes (Section VI-A), which a sum of partial errors cannot represent. Fourth, resolution is varied before segmentation in one physical frame, so the engine meets each level as an acquisition, topology and local defects are scored with geometry, and the findings are re-tested on a held-out organ. To our knowledge, no prior study combines these elements.

III. Definitions

Reference standard. The *expert mask* is a manual segmentation, a reference standard rather than the truth. The *reference surface* S_R is built from it by the *canonical* configuration: no blur, marching cubes at level 0.5 [5], no smoothing, no decimation, at native spacing. A *test mesh* S_T is the output of any configuration. In Tier 1 the exact surface replaces S_R.

Surface measures. With point sets P_T, P_R sampled uniformly by area on S_T, S_R and $d(p, S)$ the point-to-triangle distance,

$$\mathrm{ASSD} = \frac{\sum_{p\in P_T} d(p, S_R) + \sum_{q\in P_R} d(q, S_T)}{|P_T| + |P_R|}. \qquad (1)$$

HD95 and P99 are the larger of the two directed 95th and 99th percentiles, NSD_τ the share of points within τ of the other surface [14], and $\bar{d}_\pm$ the mean signed distance of P_T to S_R (negative means shrinkage). A *local defect* is a face of S_T whose vertices all lie more than 2 mm from S_R; defects sharing a vertex form a patch. A mesh is *closed* when it has no boundary and no non-manifold edges.

Added sampling. A resolution level sets a minimum spacing m_a per axis, so $s'_a = \max(s_a, m_a)$ and scans are only coarsened. Because sampling blur adds in quadrature, we define

$$\Delta = \sqrt{\tfrac{1}{3}\textstyle\sum_a \left(s'^2_a - s^2_a\right)}, \qquad (2)$$

which equals h when an isotropic grid h replaces an infinitely fine one. Attribution uses the main-effect share $\eta^2 = \sum_l n_l(\bar{y}_l - \bar{y})^2 / \sum_i (y_i - \bar{y})^2$, with the remainder assigned to interactions and residual [17]. It describes the tested grid; it is not a causal contribution.

IV. Evaluation Framework

Both tiers call the reconstruction function that the application itself uses (Fig. 1). It applies Gaussian blur to the binary mask, marching cubes at level 0.5, Taubin smoothing (pass band 0.05) [6] and topology-preserving decimation (feature angle $45°$) [7]. Research grids specify blur in millimetres, so that blur is not confounded with spacing; the application specifies it in voxels.

Tier 1. Four phantoms with exact signed distance functions covered distinct surface types: a sphere (radius 15 mm), a thin capsule (radius 2 mm, length 44 mm), a torus (radii 18 and 6 mm; genus one) and a rounded box ($28 \times 20 \times 12$ mm, 2 mm rounding). Each was placed with a random rotation and sub-voxel offset, voxelised, reconstructed and scored with exact distances in both directions, which also measures the error of the procedure that builds the Tier 2 references.

Tier 2. Each CT scan and, separately, its expert mask were resampled to each level *before* segmentation. Each coarsened axis was Gaussian filtered with $\sigma_a = (s'_a/s_a - 1)/2$ voxels, the anti-aliasing convention of scikit-image [18], then linearly interpolated so that voxel j lies at $j\, s'_a$. All grids thus share one physical frame and need no offset against S_R. Resampling the CT, rather than coarsening a finished mask, lets the engine respond to each level as it would to a coarser acquisition; it returned its mask on the same grid. The resampled expert mask, thresholded at 0.5, is a segmentation-free control that isolates sampling error. As a sensitivity analysis, coarsening was repeated with a rectangular slice profile one coarse voxel wide and, for the engine, with white noise (SD 20 HU) added to the coarse CT.

Design. The design is split-plot [17]: whole plots (one segmentation each) are mask source × resolution level, and sub-plots (one mesh each) are reconstruction settings, crossed in a full factorial grid plus the application's *production* settings and a *production-mm* variant that keeps them but sets blur to 0.5 mm, which separates blur units from the grid.

V. Data and Experimental Setup

Data. The spleen set was the 41 training scans of Medical Segmentation Decathlon (MSD) Task09 [9] (CC-BY-SA 4.0), each with one expert segmentation; in-plane spacing was 0.61–0.98 mm and slice spacing 1.5–8 mm (5 mm in 24 scans).

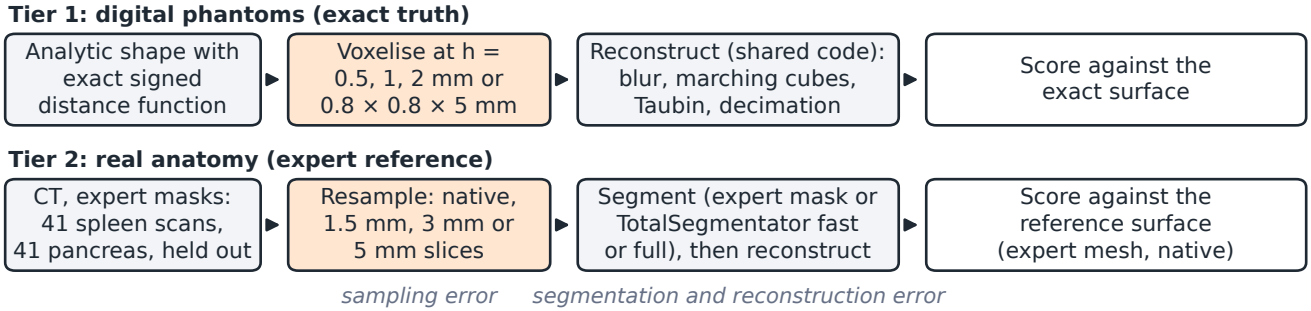


Fig. 1. Study overview. Tier 1 scores reconstructions of analytic phantoms against their exact surfaces. Tier 2 resamples expert-labelled CT before segmentation and scores meshes from the expert mask and from TotalSegmentator against the expert reference surface; the pancreas is a held-out test of the spleen findings.

The held-out set was 41 of the 281 training scans of MSD Task07 Pancreas, drawn at random (seed 2026) before any was processed; its pancreas and tumour labels were merged, as the engine's pancreas includes lesions; in-plane spacing was 0.64–0.98 mm and slice spacing 1.25–5 mm (5 mm in 6 scans). Both sets come from one centre; TotalSegmentator was trained on data from another hospital [2], so the scans are external to it, and none tuned any setting.

Factors. Tier 1 voxelised each phantom at isotropic $h \in \{0.5, 1, 2\}$ mm and at $0.8 \times 0.8 \times 5$ mm, at five placements. Its grid crossed blur $\{0, 0.5, 1\}$ mm, isosurface variant (Lewiner, Lorensen), Taubin iterations $\{0, 10, 30, 60\}$ and decimation $\{0, 0.25, 0.5, 0.75\}$ (fraction of faces removed): 96 configurations plus production (blur 0.6 voxels, 30 iterations, decimation 0.5), giving 7,760 meshes. Tier 2 fixed the isosurface variant, which explained 0.0% of Tier 1 variance, leaving 48 grid configurations, production (decimation 0.6 for the engine, as in the application) and production-mm. Its levels were `native`; `iso1.5` and `iso3` (minimum 1.5 and 3 mm on every axis); and `slice5` (minimum 5 mm slices, native in-plane). Level effects use only the scans a level resampled: all scans at `iso1.5` and `iso3`, and at `slice5` the 14 spleen and 35 pancreas scans with thinner slices. With the sensitivity runs, 57,944 meshes were scored (S1).

Engine. TotalSegmentator 2.13 segmented each resampled scan in fast mode, the application's default, which predicts at 3 mm, and in full-resolution mode, which predicts at 1.5 mm; the latter ran only the sub-model holding the organ, on a crop found by the 3 mm model, as for single-organ use. Both return the mask on the input grid.

Held-out protocol. Six hypotheses (H1–H6, Table IV) and their pass criteria were derived from the spleen results and recorded in the repository before any pancreas scan was segmented or scored; every outcome is reported.

Metrics and statistics. Each surface was sampled with 20,000 points per side; distances were exact point-to-triangle distances (VTK); $\tau = 1$ mm unless stated. Mask-level Dice compared each mask with the native expert mask after nearest-neighbour mapping to the native grid. Level effects were paired by scan, with 95% bootstrap confidence intervals over scans (5,000 resamples) [19] and Wilcoxon signed-rank tests [20] with Holm correction [21]. Variance shares are descriptive main effects across the tested grid, with 95% CIs from 1,000 bootstrap resamples of scans (Tier 2) or of placements within shape (Tier 1); a random-intercept model, estimated by ANOVA on the balanced levels (equal to REML there), instead estimates the variance between scans after removing within-scan noise. The sampling relationship was fitted through the origin, since $\Delta = 0$ reproduces the reference exactly; its 95% prediction interval is the 95th percentile of the absolute leave-one-scan-out residuals. Python 3.14 (NumPy, SciPy, scikit-image, PyVista/VTK) and an RTX 5080 GPU; seed 2026.

VI. Results

All 57,944 meshes produced a surface, and no mask was empty. Re-scoring the spleen with the extended code reproduced every value of the first version exactly.

A. Q1: Reconstruction error on exact surfaces

Spacing dominates reconstruction error. The canonical configuration, which builds the Tier 2 references, has an ASSD of about $0.14\,h$ for every shape (S2), with small placement-to-placement variation (coefficient of variation ≤ 0.19). Spacing is the largest main effect, with 38.8% [36.1, 42.0] of ASSD variance against 21.3% for shape; on the log scale, where its multiplicative interaction with shape largely disappears, it explains 68.6%. Blur, smoothing, decimation and the isosurface variant together explain 2.2% (Table III). Smoothing depends on shape: at 2 mm, 30 Taubin iterations, averaged over the other factors, reduce ASSD for the sphere (0.309 to 0.199 mm), torus and box but double it for the capsule (0.425 to 0.844 mm). At $0.8 \times 0.8 \times 5$ mm, 64% of capsule meshes split, as 5 mm slices cannot sample a 4 mm tube continuously.

B. Q2: Sampling error on real anatomy

Simulated coarsening alone moves a correct surface, predictably. Resampling the spleen's expert mask moves its canonical surface by 0.204, 0.418 and 0.445 mm at `iso1.5`, `iso3` and `slice5` (Table II; all $p < 0.001$), while Dice stays at 0.985, 0.968 and 0.971. The error follows the added sampling (Fig. 2): ASSD $= 0.177\,\Delta$, 95% CI [0.171, 0.184], $R^2 = 0.76$;

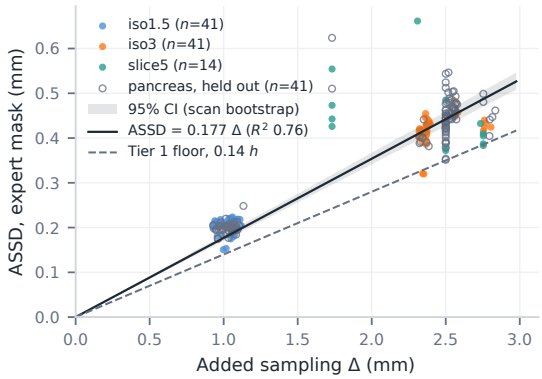


Fig. 2. Canonical ASSD of the resampled spleen expert mask against the added sampling Δ of (2): 96 points from 41 scans. Solid: fit through the origin, with its 95% CI from a scan bootstrap (band). Dashed: the Tier 1 floor, $0.14\,h$. Hollow: the held-out pancreas, which the fit did not see.

predicting each scan from the other 40 gives $R^2 = 0.75$, a mean absolute error of 0.037 mm and a 95% prediction interval of ± 0.111 mm. Fitted on two levels, it predicts the in-plane levels within 0.028–0.038 mm but slice-only coarsening less well (0.109 mm). Refinements do not generalise better (S4): a free intercept misses a held-out level by 0.28 mm, and separate in-plane and slice weights converge to equal values, supporting the pooled Δ. On the held-out pancreas the expert mask moves by 0.199, 0.456 and 0.442 mm (Dice 0.973, 0.936 and 0.942); the spleen fit predicts these within 0.030 mm on average, and the pancreas's own slope is 0.180 [0.176, 0.184].

C. Q3: Automatic segmentation across resolutions and modes

In fast mode, the engine's error does not depend on in-plane resolution down to 3 mm, rises with thick slices, and is dominated by between-scan variation. On the spleen its canonical ASSD is 0.901, 0.905 and 0.907 mm at `native`, `iso1.5` and `iso3`, with both paired CIs containing zero, and its Dice is 0.940 throughout (Table II); with 5 mm slices its ASSD rises by 0.188 mm on the same 14 scans. Scans carry 89.5% [65.0, 95.2] of its ASSD variance and resolution 2.7%, with all reconstruction settings together under 0.5% (Table III); a random-intercept model assigns 94.2% to scans.

The full-resolution mode shows that this insensitivity belongs to the fast mode, which predicts at 3 mm anyway. On native spleen scans it is more accurate (ASSD 0.790 against 0.901 mm, better on 39 of 41 scans; Dice 0.951), and it stays so at `iso1.5` and with 5 mm slices; but on 3 mm input its ASSD rises by 0.420 mm [0.247, 0.712], by more than 0.2 mm on 25 of 41 scans, and its scan share falls to 59.3% as resolution-by-scan interactions grow (S10). The pancreas is harder for both modes: in fast mode its native ASSD is 2.722 mm and Dice 0.753, full mode does not help (3.021 mm, better on 18 of 41 scans), and in-plane coarsening is no longer neutral ($+0.046$ mm at `iso1.5`, $+0.120$ mm at `iso3`). Much of its error is tumour that the engine leaves out of the organ: in one scan with a 732 cm^3 tumour, Dice is 0.16 (S3).

D. Q4: Deployed settings, geometry, topology and local defects

Production settings shrink thin and coarse-grid structures: they raise the 2 mm capsule's error from 0.331 to 1.192 mm, shrinking it by more than half its radius, and move the spleen expert mask 0.234 mm at native and 0.517 mm at `iso3`, where the 0.6-voxel blur becomes 1.8 mm; holding blur at 0.5 mm instead lowers this to 0.486 mm, and changes production results by at most 0.065 mm on expert masks and 0.14 mm on engine masks (S7).

Decimation explains at most 0.9% of error variance but drives topology: on the unsmoothed expert mesh it leaves the surface unchanged ($< 10^{-6}$ mm) in 39 of 41 spleen scans at 0.25, by first merging the coplanar patches of marching cubes, yet at 0.75 only 38–85% of meshes stay closed. Meshing the engine's output on its 3 mm working grid instead of the native grid, at identical physical blur, smoothing and decimation, changes its ASSD by only -0.017 mm [-0.032, -0.003] ($p = 0.02$) but raises closure at 0.75 decimation from 57% to 83% ($+26$ points [20, 32], $p < 0.001$; 36 of 41 scans). The one spleen scan open at every setting is cut by the field of view; without it, every undecimated mesh is closed (S6).

Global means hide local defects (S8). Faces more than 2 mm from the reference cover 8.9% of the fast engine's native spleen surface (patches up to 2,190 mm^2) and none of the resampled expert mask's; no fast-mode mesh meets an HD95 of 2 mm, while 80% stay within ± 5% of the reference volume. In the worst spleen scan the engine misses 18% of the organ in one region (S3); on the pancreas, defects cover 29% of its surface, mostly missed tumour.

E. Q5: Held-out confirmation on the pancreas

Of the six hypotheses fixed from the spleen (Table IV), four were confirmed on the pancreas, one partly and one not. The sampling relationship transferred (H1): 98% of pancreas points fall within the spleen's prediction interval. Spacing dominated the correct segmentation (H2), and scans dominated the engine (H3; 98.5% without the tumour scan above). Thick slices raised the engine's error again (H5), in 34 of 35 scans. Two spleen findings did not transfer: a 3 mm in-plane grid raised the engine's error by 0.120 mm, beyond the ± 0.10 mm margin (H4), and meshing on it, while again closing 25 points more meshes, cost 0.085 mm [0.034, 0.143] (H6). Both fit a thin, lobulated organ, whose resampled expert mask also loses more overlap at 3 mm (Dice 0.936 vs 0.968).

VII. Robustness

Coarsening model. A rectangular slice profile changed the spleen expert mask's sampling error by at most 0.003 mm (slope 0.174 against 0.177) and the fast engine's error by at most 0.020 mm; white noise of 20 HU changed it by at most 0.019 mm (S9). *Configurations.* In all 49 configurations, scans out-weighed resolution for both organs and modes, the `slice5` change was positive, and the fast engine's `iso3` change stayed within -0.059 to $+0.025$ mm on the spleen but was positive on the pancreas (up to $+0.166$ mm), so H4's failure is not

TABLE II

TIER 2 RESULTS BY ORGAN, MASK SOURCE AND RESOLUTION LEVEL (MEANS ACROSS SCANS). DICE IS AGAINST THE NATIVE EXPERT MASK, ASSD (MM) OF THE CANONICAL MESH AGAINST THE REFERENCE SURFACE; CHANGE: PAIRED CHANGE IN ASSD AGAINST NATIVE [95% SCAN-BOOTSTRAP CI], OVER THE SCANS A LEVEL RESAMPLED; CLOSED: SHARE (%) OF CLOSED PRODUCTION MESHES. MEDIANS, IQRS AND WORST CASES ARE IN S3.

		Spleen					Pancreas				
Source	Level	n	Dice	ASSD	Change vs native	Closed	n	Dice	ASSD	Change vs native	Closed
Expert mask	`native`	41	–	0.000	–	98	41	–	0.000	–	98
	`iso1.5`	41	0.985	0.204	$+0.204$ [$+0.198$, $+0.209$]	95	41	0.973	0.199	$+0.199$ [$+0.196$, $+0.203$]	95
	`iso3`	41	0.968	0.418	$+0.418$ [$+0.409$, $+0.426$]	98	41	0.936	0.456	$+0.456$ [$+0.446$, $+0.467$]	100
	`slice5`	14	0.971	0.445	$+0.445$ [$+0.410$, $+0.489$]	93	35	0.942	0.442	$+0.442$ [$+0.425$, $+0.461$]	94
TS fast	`native`	41	0.940	0.901	–	66	41	0.753	2.722	–	37
	`iso1.5`	41	0.940	0.905	$+0.004$ [-0.019, $+0.026$]	90	41	0.751	2.768	$+0.046$ [$+0.013$, $+0.083$]	85
	`iso3`	41	0.940	0.907	$+0.006$ [-0.006, $+0.018$]	95	41	0.753	2.842	$+0.120$ [$+0.072$, $+0.176$]	93
	`slice5`	14	0.926	1.079	$+0.188$ [$+0.152$, $+0.228$]	50	35	0.747	2.181	$+0.143$ [$+0.116$, $+0.170$]	54
TS full	`native`	41	0.951	0.790	–	98	41	0.738	3.021	–	76
	`iso1.5`	41	0.951	0.805	$+0.015$ [-0.002, $+0.029$]	93	41	0.737	3.092	$+0.072$ [-0.007, $+0.159$]	93
	`iso3`	41	0.932	1.210	$+0.420$ [$+0.247$, $+0.712$]	95	41	0.698	3.507	$+0.487$ [$+0.366$, $+0.647$]	90
	`slice5`	14	0.937	0.913	$+0.212$ [$+0.173$, $+0.251$]	86	35	0.731	2.396	$+0.232$ [$+0.178$, $+0.285$]	77

TABLE III

DESCRIPTIVE MAIN-EFFECT SHARES OF ASSD VARIANCE (η^2, %) ACROSS THE TESTED GRID, WITH 95% CIS FROM A BOOTSTRAP OVER PLACEMENTS WITHIN SHAPE (T1) OR OVER SCANS (T2). TS: TOTALSEGMENTATOR FAST (FULL MODE IN S10). T1 INTERACTIONS + RESIDUAL INCLUDE PLACEMENT REPLICATION (2.2%); A SMALL NEGATIVE REMAINDER ARISES WHERE `SLICE5` LACKS SOME SCANS, SO MAIN EFFECTS ARE NOT ORTHOGONAL. [a]VARIANCE BETWEEN SCANS IN A RANDOM-INTERCEPT MODEL ON THE LEVELS THAT RESAMPLE EVERY SCAN: A DIFFERENT ESTIMAND FROM η^2 (SECTION V).

Factor	T1 phantoms	Spleen expert	Spleen TS fast	Pancreas expert	Pancreas TS fast
Spacing / resolution	38.8 [36.1, 42.0]	50.5 [47.3, 55.7]	2.7 [0.2, 15.0]	74.2 [68.4, 80.7]	0.5 [0.1, 1.7]
Shape / scan	21.3 [19.0, 23.5]	10.3 [6.2, 14.0]	89.5 [65.0, 95.2]	4.1 [2.3, 6.0]	99.9 [98.1, 99.9]
Blur	0.8 [0.8, 0.9]	0.9 [0.6, 1.2]	0.0 [0.0, 0.0]	0.3 [0.2, 0.4]	0.0 [0.0, 0.0]
Taubin iterations	1.2 [1.0, 1.5]	16.0 [10.4, 23.0]	0.4 [0.1, 1.4]	6.6 [4.6, 9.2]	0.0 [0.0, 0.2]
Decimation	0.2 [0.2, 0.2]	0.4 [0.4, 0.5]	0.0 [0.0, 0.0]	0.9 [0.8, 1.0]	0.0 [0.0, 0.0]
Isosurface variant	0.0 [0.0, 0.0]	–	–	–	–
Interactions + residual	37.7 [35.6, 39.5]	21.9 [16.5, 25.5]	7.4 [2.9, 24.0]	13.9 [10.7, 16.3]	-0.4 [-1.6, 1.6]
Scans, random intercept[a]	–	13.0	94.2	4.7	99.9

TABLE IV

HELD-OUT CONFIRMATION, TS FAST. [a]CRITERIA FIXED BEFORE THE PANCREAS SCANS WERE PROCESSED: H1 MAE ≤ 0.074 MM; H2 RESOLUTION HAS THE LARGEST η^2 (%, VS SCANS); H3 SCAN η^2 (%) ABOVE RESOLUTION AND RANDOM-INTERCEPT (RI) SHARE ABOVE 50%; H4 CI OF THE ASSD CHANGE (MM) WITHIN ± 0.10; H5 CI ABOVE 0; H6 CLOSURE GAIN AT 0.75 DECIMATION WITH CI ABOVE 0 AND $|\Delta\text{ASSD}| \leq 0.05$ MM. [b]LEAVE-ONE-SCAN-OUT.

	Hypothesis[a]	Spleen	Pancreas	Held
H1	Spleen 0.177Δ fit predicts	MAE 0.037[b]	MAE 0.030	yes
H2	Spacing leads (expert)	50.5 vs 10.3	74.2 vs 4.1	yes
H3	Scans lead (engine)	89.5; RI 94.2	99.9; RI 99.9	yes
H4	`iso3` leaves engine	$+0.006$ [-0.006, 0.018]	$+0.120$ [0.072, 0.176]	no
H5	`slice5` raises engine	$+0.188$ [0.152, 0.228]	$+0.143$ [0.116, 0.170]	yes
H6	3 mm grid: closure up, ASSD same	$+26$ pts; -0.017	$+25$ pts; $+0.085$	partly

a property of one setting (S12). *The 14-scan subset.* All 14 resampled spleen scans moved in the same direction at `slice5`; leaving any one out kept the fast engine's effect within 0.175–0.195 mm, and the smallest effect detectable with 80% power was 0.063 mm (S5); the 35 pancreas scans replicate it with a detectable effect of 0.040 mm. *Case properties.* Lower organ-to-surround contrast went with higher engine error on both organs (Spearman $\rho = -0.59$ spleen, -0.49 pancreas). Measured scan properties explained little of the spleen's between-scan variance (leave-one-out $R^2 = 0.14$) but more of the pancreas's (0.59), where label volume, that is tumour burden, matters ($\rho = +0.49$; S11). The scan effect is thus partly explained on the pancreas and largely unexplained on the spleen.

VIII. DISCUSSION

Two regimes. With a correct segmentation, resolution leads and smoothing follows; with an automatic engine, between-scan variation leads, and resolution matters through slice thickness and, for a thin organ or a fine-resolution engine, through in-plane spacing too. These are descriptive shares of the tested grid, not causes; decimation, negligible for geometry, leads for topology.

What the held-out organ changed. The sampling relationship and both dominance patterns transferred; the spleen's insensitivity to 3 mm grids did not. It holds where the engine already works at 3 mm and the organ is compact, so a pipeline that resamples to 3 mm for speed should be checked per organ, and a finer engine is not uniformly better: full mode helped the spleen and not the pancreas.

The sampling relationship and the tolerance it implies. Within the tested protocol, ASSD $\approx 0.177\,\Delta$ turns a scan's

spacing into an expected error before segmentation; its slope changed by under 2% across organs and slice profiles. It also bounds the reference: the native reference surface carries about 0.4–0.5 mm of its own sampling error on 5 mm-slice scans and 0.2 mm on slices of 2.5 mm or less, so differences below this floor reflect the reference staircase, not anatomy, and a 0.5 mm NSD tolerance is uninformative on thick-slice data.

Relation to prior methodology. Additive error budgets [8] miss the third of phantom variance that lies in interactions; the engine's 0.9 mm spleen ASSD is close to the 0.8 mm median segmentation error of the printing literature [1], a different statistic, and its degradation at 5 mm but not 3 mm spleen slices matches the cadaver finding [10].

Implications. Report surface metrics and local defects alongside Dice, which stayed at or above 0.936 while ASSD rose by up to 0.456 mm; specify blur in millimetres; check closure after decimation; and mesh a fast engine's output on its working grid, after checking the accuracy cost for thin organs.

IX. Limitations and Threats to Validity

Internal. Coarsening is simulated: two slice profiles and noise bracket its effect (S9), but kernel, dose and partial-volume differences of real reconstructions are not reproduced, so the sampling relationship is probably a lower bound for real acquisitions. One annotator per scan means annotation style cannot be separated from engine error. *External.* Two abdominal organs from one centre and one engine in two modes; branching structures, other centres and engines are untested. *Construct.* Local defects and tolerance rates (S8) add detail, but no clinical tolerance is validated. *Conclusion.* The η^2 shares are descriptive, not causal; the spleen `slice5` subset has 14 scans (S5), and the held-out set tests six hypotheses once.

X. Conclusion

Reconstruction error is about $0.14\,h$ on exact surfaces (Q1), and simulated coarsening moves a correct surface by about $0.177\,\Delta$ on two organs (Q2). The engine's error is set mostly by the scan; its insensitivity to 3 mm grids holds only for the fast mode on the compact spleen, while 5 mm slices raise the error throughout (Q3). Decimation breaks closure without moving unsmoothed geometry, meshing on the engine's grid restores it, and missed tumour dominates the pancreas's local defects (Q4); four of six pre-specified hypotheses held on the held-out organ and one partly (Q5). Crossing stages against exact and expert references and testing on a held-out organ reveals interactions and limits that additive error budgets miss. As a per-mesh estimate in the application, the results held their stated coverage (S14). Future work will add real coarse acquisitions, other centres, engines, annotators and branching structures, and predict each case's error without ground truth.

Data and Code Availability

https://github.com/Organic42/MEDSYS, release `v2.0-paper`: code, per-mesh results, the held-out hypotheses, the supplement and reproduction commands. Images: Medical Segmentation Decathlon, CC-BY-SA 4.0 [9]; not redistributed.

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